

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 2.4: Surface Area and Volume of Solids — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.4: Surface Area and Volume of Solids
Driving Question	DRIVING QUESTION: How does the shape of a solid determine how much material is needed to build it and how much it can hold — and how do engineers and manufacturers in Kenya use this knowledge to design efficient, cost-effective structures?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Surface Area of Prisms and Pyramids — Launching the Phenomenon	Learners folded and unfolded pre-printed nets of a rectangular prism (a water trough), a triangular prism (a roof support beam), and a square-based pyramid (a decorative roof cap) — all modelled as flat metal sheets from the Kisumu factory. They measured and summed individual face areas to estimate total surface area before seeing any formula.	Learners derived and applied the surface area formulae for rectangular prisms ($SA = 2(lw + lh + wh)$), triangular prisms ($SA = bh + (a+b+c)l$), and square-based pyramids ($SA = l^2 + 2ls$). They connected each algebraic term to a specific face of the net, and applied formulae to calculate the metal sheet needed for factory products.	The total surface area of a prism or pyramid is the sum of the areas of all its faces, which can be found by unfolding the solid into a flat net. Different solid shapes — even if they enclose the same volume — can have very different surface areas. This is why the frustum tank in Kisumu uses less metal sheet than the cylinder-hemisphere tank despite holding the same 5,000 litres of water.
Lesson 2 Surface Area of Cones and Spheres	Learners unfolded a cone net and observed that the curved lateral surface is a sector of a circle. They measured the slant height and identified that the arc length of the sector equals the circumference of the cone's base. They also observed an orange peel laid flat over four circles of equal radius to the orange's cross-section.	The total surface area of a cone is $SA = \pi r(r + l)$, where r is the base radius and l is the slant height (not the vertical height). The surface area of a sphere is $SA = 4\pi r^2$, covering the entire curved surface with no flat face. For the same base radius and height, a cone has a smaller total surface area than a cylinder.	The cone surface area formula comes from adding the area of the circular base (πr^2) to the lateral surface area derived from the sector (πrl), giving $\pi r(r + l)$. The sphere formula ($4\pi r^2$) was derived intuitively from the orange peel activity and confirmed mathematically. These formulae explain why curved solids can enclose or cover space differently from flat-faced solids — a key piece of evidence for understanding why the Kisumu tank manufacturer can use different amounts of metal sheet for tanks of the same volume depending on their shape.
Lesson 3	Learners watched a video showing how a	Learners learned that a frustum of a cone	Learners explained why the frustum-shaped

Surface Area of Frustums of a Cone	frustum is formed by cutting a cone parallel to its base, and observed the step-by-step derivation of the curved surface area formula $\pi l(R + r)$ and the total surface area formula $SA = \pi(R^2 + r^2) + \pi(R + r)l$, pausing at three key moments to sketch, label, and record the formula.	has two circular faces (radii R and r) and a curved lateral surface, and that its total surface area is calculated using $SA = \pi(R^2 + r^2) + \pi(R + r)l$; they applied this formula to calculate that the Kisumu frustum-shaped water tank has a total surface area of approximately 7.03 m^2 , costing approximately Ksh 5,975 in metal sheet.	tank in the anchoring phenomenon uses less metal sheet than the composite cylinder-hemisphere tank by comparing their calculated surface areas, and justified — in the role of a Kisumu tank manufacturer — which shape they would choose to produce based on cost and material efficiency evidence.
Lesson 4 Surface Area of Frustums of a Pyramid	Learners watched a YouTube video on the surface area of a frustum of a pyramid, pausing to annotate a three-column observation worksheet identifying the net components (two square/rectangular faces and four trapezoidal lateral faces) and the steps used to assemble the surface area formula.	Learners derived the surface area formula for a frustum of a square pyramid — $SA = a^2 + b^2 + 2l(a + b)$ — by decomposing the solid into its net, calculating the area of each face type, and summing them. They applied this formula to calculate the sheet metal required for a frustum-shaped grain store in Kitale.	Learners explained, using their derived formula and their cumulative solid model, how the slant height and the top and bottom base lengths of a frustum of a pyramid independently determine its surface area — regardless of the volume enclosed — connecting this insight to the anchoring phenomenon of the Kisumu tank manufacturer's choice of shape to minimise material cost.
Lesson 5 Surface Area of Composite Solids	Learners watched a video demonstration of the three-step composite surface area strategy (identify components, subtract hidden faces, sum exposed surfaces) and examined a labelled diagram of a Kericho tea canister (cylinder + cone) to identify which faces are exposed and which are hidden at the joint.	Learners calculated that the cylinder-hemisphere water tank has a total exposed surface area of approximately 16.09 m^2 ($5.12\pi \text{ m}^2$), which is measurably larger than the frustum tank's surface area computed in Lessons 3–4 — even though both tanks hold exactly 5,000 litres of water. They learned that the composite surface area equals the sum of all component surface areas minus twice the area of each shared (hidden) face at a junction.	Learners explained — using their Comparison Table and annotated cumulative model — that the frustum tank uses less metal sheet than the cylinder-hemisphere tank because its shape produces a smaller total surface area for the same enclosed volume, providing direct mathematical evidence to answer the driving question about why the Kisumu factory's frustum tank is more material-efficient.
Lesson 6 Volume of Prisms, Pyramids, Cones, and Spheres	Learners observed that filling a cone with sand and pouring it into a cylinder of equal base and height requires exactly 3 cones to fill the cylinder — providing concrete, physical evidence that the volume of a cone is one-third the volume of its corresponding cylinder. They also observed labelled diagrams of two grain stores (a rectangular prism and a cone) in Kitale, Trans-Nzoia County, with identical base widths and heights.	Learners learned the volume formulae for prisms ($V = \text{Base Area} \times h$), pyramids and cones ($V = \frac{1}{3} \times \text{Base Area} \times h$), cylinders ($V = \pi r^2 h$), and spheres ($V = \frac{4}{3} \pi r^3$). They discovered that the $\frac{1}{3}$ factor arises because a cone or pyramid occupies exactly one-third of the space of a prism or cylinder with the same base and height. They learned to convert between m^3 and litres ($1 \text{ m}^3 = 1\,000 \text{ L}$) and calculated that the cylindrical-hemispherical Kisumu tank holds approximately 5 030 litres — consistent with the manufacturer's 5 000-litre claim.	Learners explained that the Kitale farmer's rectangular grain store holds approximately 3.8 times more maize than the conical store with the same base and height, because the cone's tapering shape means it occupies only one-third of an equivalent cylinder's volume. They further explained that the Kisumu tank manufacturer can engineer equal volumes (5 000 litres) into very different shapes — cylindrical-hemispherical vs. frustum — confirming that volume and surface area are mathematically independent properties, and that shape choice affects material cost (surface area) without necessarily changing storage capacity (volume).

Lesson 7 Volume of Frustums of a Cone	Learners watched a YouTube video showing how the volume of a frustum is derived by subtracting the smaller removed cone's volume from the larger original cone's volume, pausing at two designated points to record and discuss the mathematical steps.	The volume of a frustum of a cone is given by $V = (1/3)\pi h(R^2 + Rr + r^2)$, derived from the difference of two cone volumes using similar-triangle proportionality. Learners applied this formula to calculate the volume of the Kisumu factory's frustum-shaped water tank and converted the result to litres to evaluate the manufacturer's 5,000-litre storage claim.	Learners explained, in their cumulative Tank Comparison Models, how the calculated volume of the frustum tank either confirms or challenges the manufacturer's advertised capacity of 5,000 litres, and identified that the composite cylinder-hemisphere tank's volume must still be calculated before a full comparison — and a complete answer to the driving question — can be made.
Lesson 8 Volume of Frustums of a Pyramid	Learners observed a YouTube video demonstrating that a frustum of a pyramid can be formed by removing a smaller pyramid from a larger one, and that its volume is found by subtracting the two pyramid volumes — yielding the formula $V = (h/3)(A_1 + A_2 + \sqrt{A_1A_2})$. They saw this applied to frustum-shaped structures including grain stores and construction bases in Kenya.	Learners derived and applied the formula for the volume of a frustum of a pyramid: $V = (h/3)(A_1 + A_2 + \sqrt{A_1A_2})$, where A_1 and A_2 are the areas of the two parallel bases and h is the vertical height. They learned that this formula shares the same structural logic as the cone-frustum formula from Lesson 7, with circular cross-section areas (πr^2) replaced by polygonal base areas.	Learners explained that the shape of a frustum of a pyramid determines both how much it can hold (volume) and — when combined with surface area calculations — how much material is needed to build it. They connected this to the Kisumu factory phenomenon by reasoning that manufacturers choose between frustum shapes (cone-based vs. pyramid-based) based on comparative volume-to-surface-area efficiency, a question they will fully resolve in Lessons 9 and 10.
Lesson 9 Volume of Composite Solids — Solving the Kisumu Tank Mystery	Learners watched a video on composite solid volumes and observed how any composite solid can be decomposed into simple constituent shapes (e.g., cylinder + hemisphere, cone + cylinder). They observed the decomposition strategy applied step by step, and then applied it themselves to compute the volumes of both Kisumu factory tank designs.	Learners learned that: (a) the volume of a composite solid equals the sum of the volumes of its constituent parts; (b) the volume of the cylinder-hemisphere tank is $V = \pi r^2 h + (2/3)\pi r^3$; (c) the volume of the frustum tank is $V = (\pi h/3)(R^2 + Rr + r^2)$; (d) both tanks hold approximately 5,000 litres (equal volumes) but have different surface areas — confirming that volume and surface area are independent properties of a solid.	Learners explained — using a completed Comparison Table with actual computed values — that the frustum tank uses less metal sheet than the cylinder-hemisphere tank because it has a smaller surface area, even though both tanks hold exactly the same volume of water (5,000 litres). This explains the observation made by students at the Kisumu factory: shape determines how much material is needed (surface area) independently of how much water is stored (volume), and manufacturers strategically choose the frustum shape to reduce metal sheet consumption and lower production costs for rural households in Siaya County.
Lesson 10 Real-Life Applications, Final Explanation and Cumulative Model Presentation	Learners observed that three different solid shapes (a pure cylinder, a composite cylinder-hemisphere, and a frustum) can each hold exactly 5,000 litres (5 m^3) of water while having significantly different outer surface areas. By calculating and comparing the surface area and associated metal sheet cost (at KSh 850/m²) for each	Learners learned that surface area and volume are independent properties of a solid — a shape can be redesigned to hold the same volume while using less surface material by changing its geometry. They learned and applied the full set of surface area and volume formulas for cylinders, hemispheres, composite solids, and	Learners can now fully explain the anchoring phenomenon: the frustum-shaped tank at the Kisumu factory uses less metal sheet than the composite cylinder-hemisphere tank for the same 5,000-litre volume because the frustum's geometry — specifically its sloping lateral surface, defined by slant height $l = \sqrt{h^2 +$

	design in the AquaFab Industries engineering task, learners directly observed that the frustum design produces the lowest surface area — and therefore the lowest manufacturing cost — for the same storage volume. They also observed, through the Gallery Walk, that different groups reached the same mathematical conclusion via different calculation pathways.	frustums in a multi-step real-world engineering context. They learned that manufacturers and engineers in Kenya strategically select solid shapes to minimise material cost while meeting capacity requirements, and they connected this mathematics to careers in water engineering, architecture, packaging design, and agricultural storage across Kenya.	$(R-r)^2]$, and the fact that its top opening is narrower (smaller r) than a full cylinder's top — results in a smaller total outer surface area. Volume and surface area are determined by different aspects of a solid's shape and dimensions; they do not change proportionally together. Engineers exploit this geometric independence to design cost-efficient, high-capacity storage structures.
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